

2023

HL Mando

Developing Connected Vehicle Software Using AWS IoT Services with HL Mando

Learn how automotive parts and software supplier HL Mando facilitates the development of software-defined vehicles using AWS IoT services.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

Achieves

secure vehicle connectivity

Scales

automatically

Efficiently

collects vehicle data from standard ser

Enhances

driver convenience and vehicle performance



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Overview

Software innovations are transforming the automotive industry, fueling new in-vehicle experiences and improving driver comfort and convenience. Automotive parts and software supplier [HL Mando](#) recognized the demand for software solutions and has since specialized in delivering connected experiences to both manufacturers and their customers. Because newer car models feature more complex systems and richer sensors, HL Mando needed to continue advancing its data collection and storage to deliver more intelligent software features and upgrades.

To advance its technology offering, HL Mando built the Mando Integrated Control Software Assembly (MiCOSA) solution using Amazon Web Services (AWS) Internet of Things (IoT) services. Through MiCOSA, HL Mando has developed mobility features that provide predictive maintenance recommendations, driver alerts, and a more personalized driving experience.



Opportunity | Using AWS IoT FleetWise to Collect and Store Vehicle Data in the Cloud for HL Mando

For over 60 years, HL Mando has pioneered vehicle solutions that help improve safety, promote sustainability, and simplify mobility. Headquartered in Korea, the company has a history of manufacturing chassis and chassis components for automakers and other automotive companies. As automotive technology has advanced, HL Mando has focused on developing electric vehicle and cross-domain solutions between chassis systems and advanced driver assistance systems (ADAS). Its solutions provide connected experiences through a software-as-a-service business model.

In 2021, HL Mando accelerated its pivot toward software solutions by forming the Software (S) Laboratory, a division that is responsible for building advanced software solutions. One of the projects was MiCOSA, a connected software solution that would serve as the foundation for c



related data and launching new software products. “Before MiCOSA, we developed only chassis-related products,” says Jihyun Jung, head of the SW Beacon Laboratory at HL Mando. “As we worked to expand our features, especially for chassis and ADAS cross-domain features, we needed to collect, integrate, and store data from multiple systems inside of vehicles.”

The HL Mando team realized it needed a tech stack that could collect data from vehicle sensors that extended beyond the chassis. So, the company created proofs of concept for a handful of different IoT solutions, including the use of [AWS IoT FleetWise](#), which helps automotive companies efficiently collect vehicle data and organize it in the cloud. “AWS offers IoT solutions that are purpose-built for vehicles,” says Dr. Sangjin Ko, senior engineer at HL Mando. “AWS IoT FleetWise resolved the challenges we faced with collecting, storing, and transferring data. It is efficient and reliable.”

Solution | Developing Features to Enhance Vehicle Performance and Driver Convenience on AWS

HL Mando worked alongside the AWS team to develop MiCOSA, which includes all technology related to a vehicle’s longitudinal and lateral movement and vertical dynamics by collecting data from vehicle chassis systems and sensors. HL Mando offers MiCOSA to automakers, giving them the ability to activate software-based vehicle features for their customers. Manufacturers can use all of MiCOSA’s feature modules, or they can select individual modules according to their needs.

To establish secure vehicle connectivity on the backend of MiCOSA, HL Mando uses [AWS IoT Core](#), a service that can connect billions of IoT devices and route trillions of messages to AWS services without having to manage infrastructure. To remotely run software updates, the company uses [AWS IoT Greengrass](#), an open-source edge runtime and cloud service for building, deploying, and managing device software.

Initially, HL Mando had issues with collecting vision system data, such as data from vehicle cameras. However, the company was able to collect vision system data with MiCOSA by adopting the [AWS IoT FleetWise vision system data collection feature](#). “The AWS solutions architects gladly guided us through how to address and fix the issue,” says Dr. Ko. “We had a technical cadence meeting every week, which was a great help.”

By adopting AWS IoT services, HL Mando accelerated the development time for MiCOSA. “We have done virtually all the undifferentiated work, empowering us to focus on feature differentiation, which is what really



matters,” says Dr. Ko. As a result, HL Mando’s software division has developed several MiCOSA modules, including Prognostics and Health Management (PHM), Road Surface Perception (RSP), and Intersection Driver Assistance Warning (iDAW).

Using the PHM feature, drivers can benefit from prescriptive maintenance recommendations and analytics that are powered by deep learning algorithms. Service centers and repair shops also can use this feature to detect potential vehicle issues more quickly, despite increased software complexity. “If there’s a future failure in one of our products, PHM will detect it and suggest the necessary maintenance,” says Dr. Ko.

To improve vehicle stability, the RSP feature uses data from vehicle cameras to assess road conditions and help drivers optimize their use of chassis and ADAS-related features, such as adaptive vehicle traction control and automatic braking. With the iDAW feature, drivers can get alerts when traffic signals change to green and it’s time to go. “The time that it takes to detect changes in road conditions and send alerts is less than 1 second,” says Dr. Ko. “It’s amazing.”

Outcome | Deploying More ADAS and Electric Vehicle Software Solutions with Cloud Connectivity

With a strong foundation for cloud connectivity, HL Mando will continue expanding its feature sets for ADAS and electric vehicles. “We believe that our future success depends on cloud-level connectivity,” says Jung. “Using AWS IoT services has indeed been a game changer for us because it gives us the ability to develop in-vehicle applications rapidly and at scale.”

About HL Mando

HL Mando is a Korean automotive parts manufacturer that specializes in the development of advanced driver assistance systems and electric vehicle software solutions.

AWS Services Used

AWS IoT FleetWise

AWS IoT FleetWise makes it easier for you to efficiently collect vehicle data and organize it in the cloud so you can use the data to improve vehicle quality, performance, and autonomy.

[Learn more »](#)

AWS IoT Core



AWS IoT Core lets you connect billions of IoT devices and route trillions of messages to AWS services without managing infrastructure.

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AWS IoT Greengrass

AWS IoT Greengrass is an open-source edge runtime and cloud service for building, deploying, and managing device software.

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JAPAN



Hitachi on AWS

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KOREA

SINGAPORE





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1 2 3 4 5 6 ... 44

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2

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